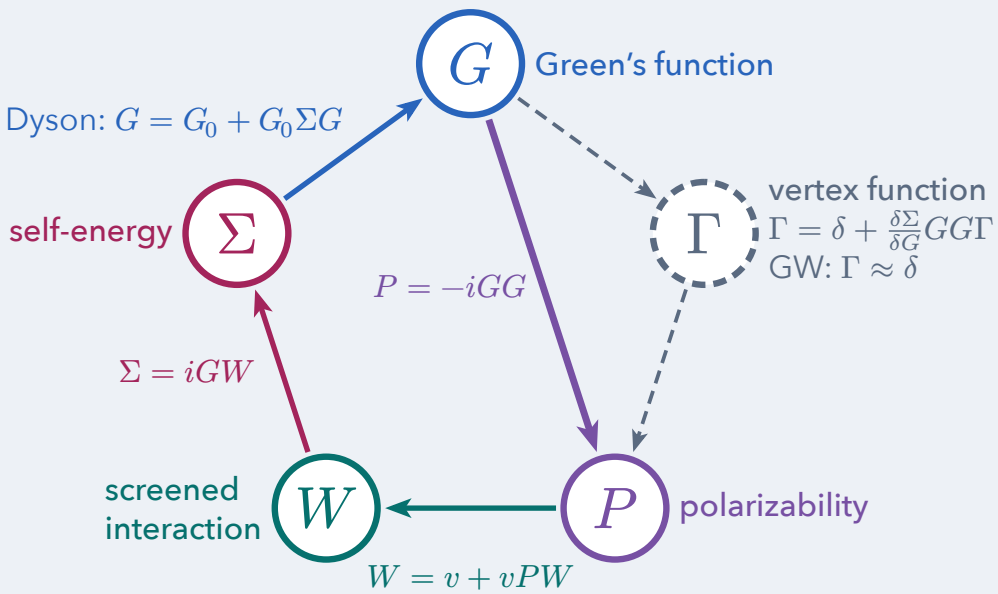
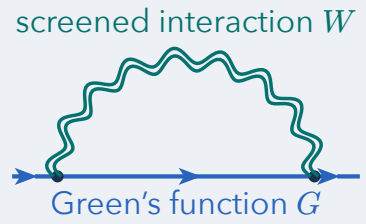
1 Hedin's equations and the GW shortcut



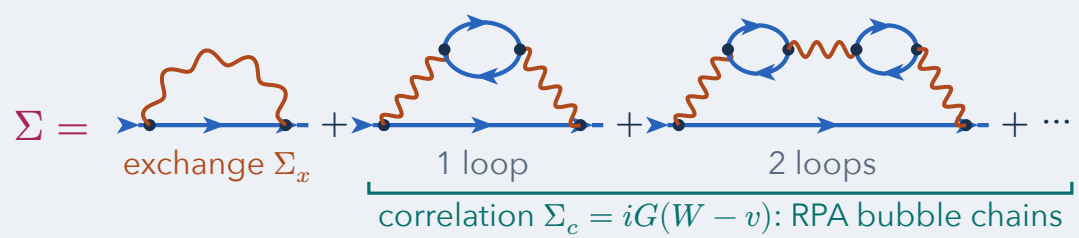
Exact theory: $P = -iGG\Gamma$ and $\Sigma = iGW\Gamma$. GW drops the vertex ($\Gamma = \delta$), closing the loop $G \to P \to W \to \Sigma \to G$. With $P = -iGG = \chi_0$, the independent-particle response, W is the RPA screening of the bare Coulomb interaction v. One-shot G_0W_0 goes round once starting from DFT orbitals and energies; GW_0, evGW and QSGW iterate.

2 The GW self-energy



self-energy $\Sigma = iGW$

An added electron or hole drags a screening cloud along: it propagates with the Green's function G and interacts with itself via the screened Coulomb interaction W, not the bare Coulomb interaction v.

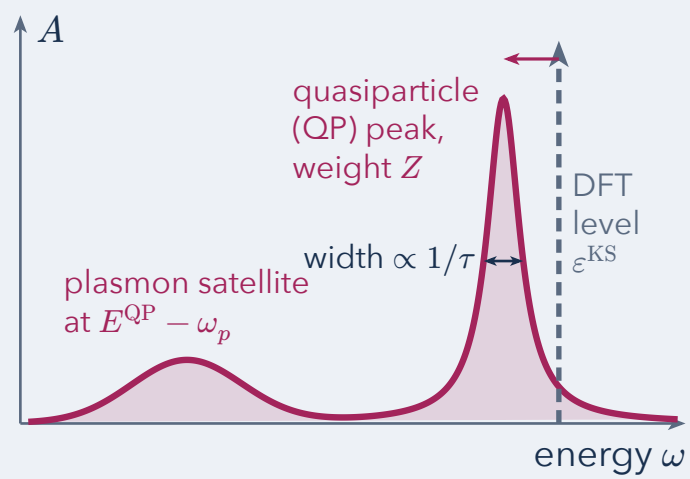


Hartree-Fock is GW with W replaced by v. GW sums every bubble chain but no crossed diagrams; the first it misses, second-order screened exchange, is the leading vertex correction. Dynamic screening makes $\Sigma(\boldsymbol{r}, \boldsymbol{r}', \omega)$ non-local, energy dependent and complex; $\mathrm{Im}\ \Sigma$ sets lifetimes.

3 Quasiparticle peak

spectral function $A(\boldsymbol{k}, \omega) = \frac{1}{\pi}\ |\mathrm{Im}\ G(\boldsymbol{k}, \omega)|$

Photoemission removes an electron from an occupied state and the other electrons screen the hole. The real part of the self-energy Σ shifts the peak away from the DFT level, its imaginary part broadens it (finite lifetime τ), and exciting plasmons moves weight into satellites.

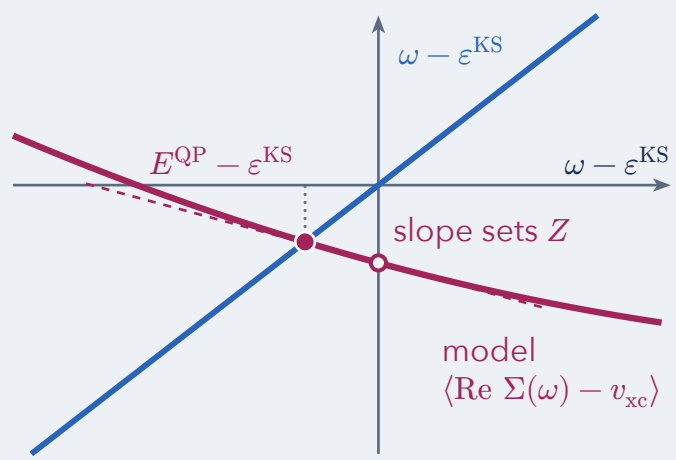


weight $Z = (1 - \partial\ \mathrm{Re}\ \Sigma/\partial\omega)^{-1} \approx 0.7\text{–}0.8$
QP energy E^{QP}, plasma frequency ω_p

4 Quasiparticle equation

$$E^{\mathrm{QP}} = \varepsilon^{\mathrm{KS}} + \langle\psi \mid \Sigma(E^{\mathrm{QP}}) - v_{\mathrm{xc}} \mid \psi\rangle$$
$$\approx \varepsilon^{\mathrm{KS}} + Z\langle\psi \mid \Sigma(\varepsilon^{\mathrm{KS}}) - v_{\mathrm{xc}} \mid \psi\rangle$$

Each Kohn-Sham eigenvalue $\varepsilon^{\mathrm{KS}}$ (orbital ψ) is corrected by swapping the DFT exchange-correlation potential v_{xc} for the self-energy Σ, evaluated at the quasiparticle energy E^{QP} itself: the solution is where the line meets the curve. Linearizing at $\varepsilon^{\mathrm{KS}}$ (dashed tangent) with the quasiparticle weight Z skips the root search.



5 Band gap of Si

Kohn-Sham eigenvalues are not electron addition and removal energies. GW supplies the missing self-energy and opens the gap between valence and conduction bands close to experiment. Valence band maxima (VBM) aligned.

